The food model, worked example (specification version 4, M101)

City E: $N = 100{,}000$; $\mathbf{w} = (0.2,\ 0.4,\ 0.4)$ for vigorous, active and sedentary work. Values are computed by `plan/tools/food_model.py` from the data files and rounded for display.

Availability (1) and the per-good data (section 3a)

g	S_g **[t]**	a_g **[g/(p·d)]**	u_g	e_g **[kcal/g]**	p_g **[g/g]**	f_g **[g/g]**	σ_g	**in $\mathcal{W}$**
Cow	20,000	547.9	547.9	1.500	0.185	0.079	1.76	yes
Dairy	150,000	4,109.6	3,000	0.610	0.033	0.033	0.78	yes
Potato	60,000	1,643.8	1,643.8	0.670	0.016	0.001	3.23	no
Wheat	40,000	1,095.9	1,095.9	2.672	0.098	0.018	1.57	no
Apple	20,000	547.9	547.9	0.480	0.001	0.003	1.97	yes
Sugar	8,000	219.2	219.2	3.730	0	0	1.00	yes
Wine	15,000	411.0	411.0	0.680	0	0	0.70	yes
Tea	400	11.0	–	–	–	–	–	stage 0
Salt	500	13.7	13.7	0	0	0	–	no

Stage 0 and the needs, (2)-(4)

$$q_{\text{tea}} = \min(11.0,\ 600/8.6) = 11.0 \text{ g}; \qquad c = 11.0 \cdot 8.6/600 = 0.157$$

$$E^{\text{need}} = 1536\,(0.2 \cdot 2.25 + 0.4 \cdot 1.76 + 0.4 \cdot 1.53) + 150 \cdot 0.157 = 1536 \cdot 1.766 + 23.6 = 2{,}735 \text{ kcal}$$

$$E^{\max} = 1536 \cdot 2.40 + 23.6 = 3{,}709 \text{ kcal}$$

$$r_{\text{fullness}} = 1.36 \cdot 2{,}735 = 3{,}720; \qquad F^{\max} = 1.36 \cdot 3{,}709 = 5{,}044; \qquad P^{\max} = 228 \text{ g}$$

The program (10)-(12)

Stage	Objective	Optimum
1	$\max A$	$A^* = 1.000$: all 15 components reach their need
2	$\max \Psi$ with $A \geq 1$	$\Psi^* = 9.65$; the energy ceiling $E^{\max}$ binds
3	$\min \sum_g q_g/a_g$ with $A \geq 1,\ \Psi \geq 9.65$	the diet below

g	q_g^* **[g/(p·d)]**	$e_g q_g^*$ **[kcal]**	$\ln(1 + e_g q_g^*/100)$	$\lambda_g^{(2)}$ **[well-being per kg]**
Cow	547.9 (whole stock)	822	2.22	0.76
Dairy	2,146.0	1,309	2.65	0

g	q_g^* **[g/(p·d)]**	$e_g q_g^*$ **[kcal]**	$\ln(1 + e_g q_g^*/100)$	$\lambda_g^{(2)}$ **[well-being per kg]**
Potato	351.3	235	– (staple)	0
Wheat	0	0	– (staple)	0
Apple	547.9 (whole stock)	263	1.29	1.01
Sugar	214.5	800	2.20	0
Wine	411.0 (whole stock)	279	1.33	1.14
Σ		**3,709** $= E^{\max}$	**9.69** (the linear program's piecewise value: $9.65 = \Psi^*$)	

How the program reads:

- **Stage 1** needs only part of these goods to reach $A = 1$.
- **Stage 2** then spends the remaining room up to $E^{\max}$ on the goods in $\mathcal{W}$. Because of the logarithm, it spreads the room across all of them; beef, apples and wine run out of stock.
- **The positive $\lambda^{(2)}$:** for these three goods, more stock would raise well-being.
- **Potatoes** (351 g) are needed only to keep fullness at its need while the energy room goes to well-being goods. Wheat is not needed.
- **Stage-1 shadow values** are 0 for every good: no stock limits nutrition here.

Outcomes (13)-(17)

(13) $x = 3{,}709/2{,}735 = 1.36$; $fed = 4{,}735/3{,}720 = 1.27$; $A^* = 1$; $\Psi^* = 9.65$

(14) $harm_{\text{alcohol}} = 411.0 \cdot 0.106 = 43.6$ g alcohol/(p\cdotpd)

(15) $W^{\text{in}} = \underbrace{1{,}058}_{\text{food water}} + \underbrace{2{,}146 \cdot 1.5}_{\text{milk}} + \underbrace{411 \cdot 1.0}_{\text{wine}} + \underbrace{11.0 \cdot 43}_{\text{tea}} = 5{,}159$ ml; $W^{\text{plain}} = 0$

(16) $work = \min(1,\ 1.36) = 1$

Coverage c_i:

fullness	**energy**	**protein**	**C**	**A**	**B1**	**B3**	**folate**	**B12**	**Fe**	**Zn**	**Ca**	**I**	**K**	**Na**
1.27	1.36	3.89	1.10	3.19	2.52	2.11	1.00	8.40	2.65	3.18	3.64	2.31	2.02	2.25

Stocks (17), in tonnes:

g	**Cow**	**Dairy**	**Potato**	**Wheat**	**Apple**	**Sugar**	**Wine**	**Salt**
U_g	20,000	78,328	12,823	0	20,000	7,828	15,000	0
L_g	0	71,672	47,177	40,000	0	172	0	500
D_g	0	71,672	23,588	2,000	0	17	0	50

Points for decision

1. **Harmful and sweet goods are eaten up to their stock.** Harm is not a motive (P5), so with 411 g of wine per person available, all of it is drunk (43.6 g alcohol a day); sugar reaches 214 g a day. If people should hold back, a per-good ceiling u_g for wine, beer and sugar is the one-parameter way (values to source).
2. **The harm coefficients** that turn (14) into a health burden, and the data for tobacco.
3. **Prices:** how the shadow values (18) become gold prices in the goods economy.